

Harry Wright | Portfolio

Researcher in Precision Agriculture and Hydroponics with a Focus on Engineering Applications



"This portfolio serves as a comprehensive showcase of my academic journey and hands-on projects, reflecting my dedication to applying cutting-edge technology, sustainable practices, and research-driven solutions in real-world contexts. Here, you'll find an array of my work, including my academic CV, detailed research papers, course transcripts, and letters of recommendation, each underscoring my commitment to excellence in both theory and practice."

Project Portfolio

-  [DIY Deep Water Culture Hydroponic Bubblers for Tomato Cultivation](#) Developed a custom hydroponic system for growing tomatoes, overcoming challenges like algae growth, pH fluctuations, and root rot by iterating on aeration, nutrient adjustments, and acid buffering. Scaled the system from a single bucket to a multi-bucket setup, ensuring stable pH, EC, and optimal root growth.
-  [Multi-Sensor Water Quality Monitor for Hydroponics](#) Designed a cost-effective water quality monitoring system, integrating analog sensors to measure temperature, EC, and pH. Focused on low-cost components and precision calibration to ensure reliable, real-time measurements for hydroponic systems.
-  [Max Security Linux: OS Hardening & Network Perimeter Defense](#) Built a full-stack Linux security suite for hardening Ubuntu, implementing BIOS-level protections, network defense strategies with OPNsense, and real-time traffic inspection using Suricata IPS. Emphasized the balance between strong security and system usability.
-  [Custom 3D CAD Designs for Large-Scale Hydroponic Systems](#) Engineered 3D models for hydroponic systems using parametric design principles in OpenSCAD. Focused on optimizing root spacing, material tolerances, and structural symmetry, resulting in production-ready prototypes for 15L flood-and-drain hydroponic pots.

Academic Writing

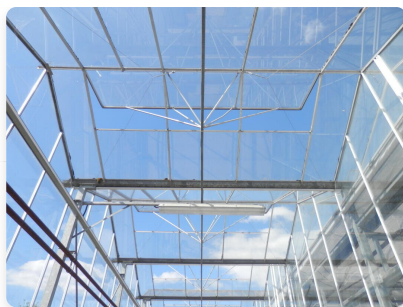
- [!\[\]\(95b42f0077faf7439a26242a54e021ec_img.jpg\) **Smart Subsurface Drip Irrigation System Design**](#) Explored smart irrigation for sustainable agriculture, specifically designing a custom ET-based scheduling model for subsurface drip irrigation (SDI) to optimize water use and crop productivity in open-field tomato farming.
- [!\[\]\(e097ab4c08b8186dd0908330bbc2dc28_img.jpg\) **Leveraging Machine Learning on CEA Sensor Data for Accurate Tomato Yield Prediction**](#) Investigated the use of machine learning (ML) to predict tomato yields based on environmental data in Controlled Environment Agriculture (CEA). The study evaluated various regression models, with Gaussian Process Regressor (GPR) showing the most promise despite data limitations.
- [!\[\]\(1e9d865c5de095f8e3304757c49e79d7_img.jpg\) **Net-Zero Agriculture: Supporting the UK's Greenhouse Gas Emissions Target**](#) Investigated strategies to support the UK's goal of achieving net-zero greenhouse gas emissions in agriculture by 2050, with a focus on practical, product-based interventions. Analyzed selected proposals from the UK Government's Carbon Budget Delivery Plan, highlighting technologies and products aligned with climate goals and sustainability.
- [!\[\]\(735b10d724a5f0ec5005c4eb3eb9c9d1_img.jpg\) **Dynamic Nutrient Dosing for Enhanced Yield in Hydroponic Strawberry Production**](#) Investigated whether dynamic, growth-stage-specific nutrient dosing improves strawberry yield in hydroponic systems compared to conventional two-phase fertilisation. Using a simulated data study informed by deep reinforcement learning principles, the research evaluated yield and nutrient efficiency across phenological stages, highlighting the potential of precision fertigation to enhance productivity while reducing nutrient waste.

Open Source

- [stm32-rs/stm32f4xx-hal \(#859\)](#)

Gallery





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